

Risk-Aware Reinforcement Learning with Group Opinion for Autonomous Driving

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Abstract—To avoid dangerous situations, such as collisions in dynamic environments, autonomous vehicles must predict the risks of the current scene to take safe actions. Traditional rule-based risk prediction methods and existing reinforcement learning (RL) approaches, which typically rely on manually designed driving decision rules or heuristic reward functions, often fail to capture the complexity of real-world dangerous scenarios, leading to suboptimal and unsafe driving decisions. To address this limitation, we develop a novel RL method, called Group Opinion Risk-Aware Reinforcement Learning (GORA-RL), for safer driving decisions that align with real-world conditions. Specifically, we first introduce surveys of human drivers to assess risk in real-world driving situations. Using these real group opinions as training data, we train a risk prediction model, referred to as the risk prediction model with a Transformer (RPT), that captures the crucial characteristics of these scenarios, resulting in more realistic and reliable risk predictions. This model is then integrated as a reward function to train an RL algorithm for making driving decisions in various scenarios. The experiments validate that our approach outperforms two state-of-the-art (SOTA) methods in challenging congested scenarios, such as merging and intersections, in terms of reward and several other metrics. Project site: <https://github.com/naiyisiji/RPT>.

I. INTRODUCTION

Autonomous driving has emerged as one of the most transformative technologies in recent years, promising to revolutionize transportation by enhancing safety, efficiency, and accessibility [1]. However, despite significant advancements, ensuring the safety and reliability of autonomous vehicles (AVs) remains a critical challenge [2]. A key aspect of this challenge is the accurate prediction and management of risks in complex driving environments [3], where the behavior of other vehicles and pedestrians can be highly unpredictable.

Current approaches to achieving safe driving can be broadly categorized into three types. The first type is traditional rule-based methods, which suffer from inefficiency due to complex scenarios and rule design. The second category includes imitation learning methods, such as [4], [5], [6], [7], [8], [9], where human demonstrations are used to train a vehicle's decision-making model. For instance, to incorporate safety constraints, prior work [10] uses predefined safety rules and expert demonstrations to ensure the autonomous vehicle operates within safe boundaries. However, limited by the given expert examples, these imitation learning methods may not fully capture the nuanced, context-dependent nature

of risk, thus limiting their ability to generalize to diverse driving conditions.

The final category includes RL methods, which are among the promising approaches and are the focus of this paper. These methods often use risk prediction or penalty functions to minimize the likelihood of collisions or other risky outcomes, requiring little manual design while autonomously making risk-aware driving decisions in various complex scenarios [11], [12], [13], [14], [15]. For example, [11] integrates a risk-based reward function into the Deep Q Network (DQN) approach, achieving better performance than conventional collision-based rewarding schemes and a rule-based intersection navigation policy. However, existing risk-aware RL methods rely on the designer's personal experience or explicit heuristic rules to design rewards, lacking diverse real-world driver opinions. As a result, these methods fail to reflect the true risk assessments of individual drivers in real-world conditions, leading to suboptimal strategies and unsafe driving decisions.

Motivated by these observations, we propose a GORA-RL approach to achieve safer driving decisions by integrating human perception of risk into the decision-making process of RL, leveraging large-scale statistical data from a diverse group of drivers. Specifically, to overcome the limitations of heuristic risk perceptions, we conduct a comprehensive survey with over 60 professional driver participants, who provide scores for various risk levels on a scale from 0 to 90 and indicate their preferred actions in different driving scenarios. The data from this comprehensive survey is then used to train a risk prediction neural network using a transformer architecture. This network takes critical driving parameters as input, including the ego vehicle's velocity and position, the relative distance and velocity of neighboring vehicles, and the distance and relative position of pedestrians. This prediction network considers group opinions and large-scale statistical data to reflect group perspectives on risk, providing a more comprehensive and realistic approach to risk prediction. The trained prediction neural network is then integrated as a risk-enhanced reward function to train a Proximal Policy Optimization (PPO) algorithm, one of the most commonly used RL algorithms in autonomous driving scenarios, to make driving decisions. This reward function maps each predicted risk level to a corresponding reward value, allowing the RL agent to make safer driving decisions across diverse scenarios.

Our main contributions are: (1) We propose a GORA-RL method for AVs, introducing group-based human risk perception into AV decision-making using large-scale sta-

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tistical survey data; (2) We develop an RPT, leveraging group opinions and key driving parameters for accurate risk prediction, and integrate it as a group opinion risk-enhanced reward function for RL; and (3) We validate our method across four key driving scenarios—loop, straight, merge, and intersection—and demonstrate significant improvements in safety and performance compared to two SOTA RL methods.

II. RELATED WORKS

Risk Prediction for AV: Risk prediction in AVs becomes a crucial research area, especially for improving safety and decision-making in dynamic environments. Traditional risk assessment often relies on rule-based systems or predefined models using sensor data to predict collisions and unsafe scenarios [16]. Another approach, such as [17], uses deep learning models to predict potential collisions based on current vehicle dynamics, road conditions, and interactions with other road users to avoid driving risks. Recent advancements directly incorporate deep learning models to estimate risk from complex environmental inputs [18], [19]. For example, convolutional neural networks (CNNs) and recurrent neural networks (RNNs) are used in [18] to predict risk levels by analyzing sensor data like LiDAR and camera feeds. However, these methods often lack the ability to integrate human-like risk perception, which is vital for safe decision-making in unpredictable environments. Our work fills this gap by incorporating group-level human opinions to provide a more comprehensive understanding of risk, ultimately enhancing AV safety.

Risk-aware RL for AV: Risk-aware RL has emerged as a promising method to incorporate safety rules or risk predictions to improve autonomous driving decisions. Most existing risk-aware RL approaches focus on introducing heuristic safety rules or constraints into the reward function to penalize unsafe actions, ensuring the agent learns to avoid high-risk situations [11], [15], [13], [20], [21], [22]. For example, safe RL algorithms have been used to ensure collision-free paths by introducing cost functions that heavily penalize any unsafe or risky behavior, guiding the vehicle towards safer trajectories [11], [13]. Recent work [4] also explores using the artificial potential field (APF) to generate risk levels and then switches between curriculum learning and human-guided learning to achieve safe, robust, and efficient RL. However, current risk-aware RL approaches rely on individual expert demonstrations or static rule-based penalties [4], which fail to capture the variability in human risk perception across different scenarios and drivers, limiting their ability to make effective decisions in complex driving environments. In contrast, our GORA-RL method integrates group opinions learned from large-scale real-world statistical data on human risk perception, resulting in more robust and safe driving decisions.

III. METHODOLOGY

This section first introduces a group opinion survey that collects risk assessment levels and preferred decisions from multiple expert human drivers for various driving scenarios.

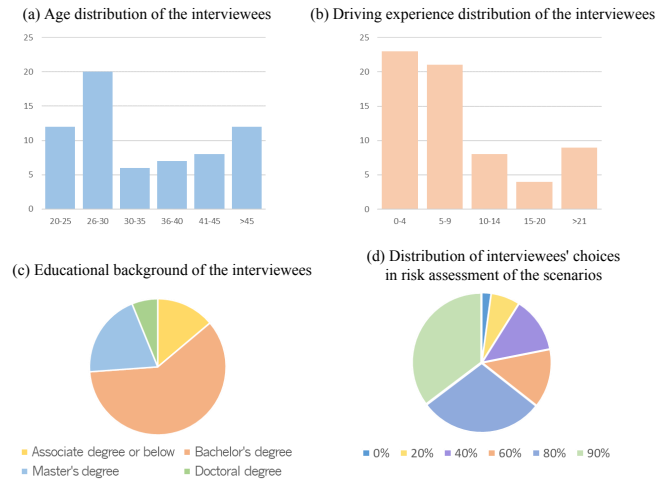


Fig. 1: Distributions of the participants by (a) age, (b) driving experience, (c) education background and (d) response of the risk level.

These assessments are used to train a risk evaluation model, which is then employed as a reward function to train the RL for driving decision-making.

A. Group Opinion Survey

To accurately assess scene risk levels in a manner consistent with real-world conditions, we designed a survey aimed at collecting human drivers' evaluations of risk levels across various driving scenarios. The images and key contextual information used in this survey were derived from the NuScenes dataset. The survey consisted of 60 questions, structured in pairs. Each pair presented participants with an image captured from the perspective of an autonomous vehicle's front camera in a dynamic scene, asking them to evaluate the risk level and suggest the appropriate action a driver should take in that scenario.

The risk level options represented the probability of a dangerous event occurring, with choices ranging from greater than 90%, 80%, 60%, 40%, 20%, to approximately 0%. This design was motivated by two considerations: firstly, to ensure that the options were clearly distinguishable without making the differences too broad; and secondly, to avoid overly fine distinctions that could result in a dispersed distribution of survey responses, potentially complicating the convergence of the corresponding risk prediction model. The highest risk level was labeled as "greater than 90%" instead of 100%, acknowledging that drivers seldom perceive any scenario as entirely certain to be dangerous. Similarly, the lowest level was designated as "approximately 0%" rather than 0% to minimize any potential bias in participants' judgments.

The survey gathered responses from 65 participants, encompassing 39 male and 26 female drivers. Participants' ages ranged from 20 to 65 years, with driving experience spanning from less than 1 year to 38 years. The educational backgrounds of participants varied from education below bachelor degrees to PhDs. Each scenario was also associated with key contextual information, including the autonomous

TABLE I: Comparison of questionnaire designs used by RPT and those used in other works.

	Participant number	Number of questions	Scenes number	Participant age range (years)	Education level coverage	Driving experience coverage (years)	Types of questions	Types of participant responses
RLHR[23]	69	15 - 25	1	Unspecified	Unspecified	Unspecified	text	multiple choice
DPQ[24]	88	63	9	Unspecified	Unspecified	Unspecified	text	multiple choice
SAFE[25]	22	22	Unspecified	56 to 88	Unspecified	Unspecified	text	Blank fill
RPT(ours)	65	64	30	20 to 65	From associate degree to PhD	less than 1 to 38	Scene photos	Blank fill and multiple choice

vehicle's speed, details of other vehicles present in the scene (e.g., relative distance and speed), and pedestrian data (e.g., the number of pedestrians and their relative distances). The distributions of the participants and response of the risk are illustrated in Fig. 1.

To validate the rationality of our questionnaire design, we compared it with the survey methods used in three similar studies, as shown in Table I. The number of questions in the questionnaire used by RLHR [23] is 15–25, which means that the number of questions to be answered may vary between 15 and 25 depending on different participants. The results indicate that the number of participants in our RPT-based survey is similar to that of other methods. However, our questionnaire provides more detailed information on participants' driving experience, educational background, and other attributes. It also includes both text and real-world images, offering more comprehensive insights than methods relying solely on text. Compared to the RLHR survey, ours covers more questions and scenarios, resulting in a richer dataset. It also addresses a broader range of scenarios and offers a more thorough background investigation than the DPQ[24] survey. Lastly, our questionnaire surpasses the SAFE[25] survey in all aspects. Thus, our participant sample size is reasonable, and our questionnaire design is superior.

We compile drivers' perceived risk levels for each scenario into a 6-dimensional vector corresponding to six risk levels set in the questionnaire, ranging from 90% to 0%. Each element of the vector represents the distribution of all participants' selections for a specific risk level. Key information is extracted from each scenario in the NuScenes dataset, including the total number of vehicles, the total number of pedestrians, the speed of the ego vehicle, the relative positions, relative speeds, and relative distances of the three nearest neighboring vehicles within a 150-meter radius of the ego vehicle, as well as the relative positions and distances of the three nearest pedestrians within the same range. By combining the key data from each scenario with the corresponding risk level vector, we create a dataset that encapsulates drivers' risk assessments for various scenarios.

B. Risk Prediction with Transformer

To model the risk assessments provided by human drivers for each real-world scenario collected through the survey, we employed a RPT that uses key contextual information associated with each scenario to predict risk levels. The

contextual information fed into the model is categorized into three types: ego features, neighboring vehicle features, and neighboring pedestrian features. The overall framework of RPT is illustrated in Fig. 2.

The ego features include the ego vehicle's speed, the number of other vehicles in the scene, and the number of pedestrians in the scene. For neighboring vehicles and pedestrians, we considered the three closest vehicles and three closest pedestrians within a 150-meter radius from the ego vehicle. Vehicles and pedestrians beyond this range have a minimal impact on the ego vehicle's decision-making process, and within this range, the motion patterns of the vehicles and pedestrians are generally comparable. Thus, selecting three neighboring vehicles and pedestrians is empirically optimal. The neighboring vehicle features include the relative speed, relative position, and relative distance with respect to the ego vehicle. The neighboring pedestrian features include the relative position and relative distance with respect to the ego vehicle. The following sections will provide a detailed explanation of the functionality implemented in each module.

1) *Adj-vehicles embedding and Adj-pedestrians embedding*: These embeddings take adj-vehicle features and adj-pedestrian features as input, respectively. The Adj-vehicle embedding process takes the features of the adjacent vehicle $\mathbf{Av} \in \mathbb{R}^{3 \times 5}$, each adjacent vehicle feature denoted as $\mathbf{Av}_i \in \mathbb{R}^5$ where $i \in \{0, 1, 2\}$, as input. The resulting embedding is denoted as $\mathbf{Bv}_i$, and the process is as follows:

$$\mathbf{Bv}_i = \mathbf{W}_{ad}(\mathbf{Av}_i) + \mathbf{W}_{as}(\mathbf{Av}_i) \quad (1)$$

where $\mathbf{W}_{ad}$ and $\mathbf{W}_{as}$ correspond to multiple and single linear layer, respectively, to capture the "deep" and "shallow" features of $\mathbf{Av}_i$. The dimension of the generated $\mathbf{Bv}_i \in \mathbb{R}^h$, where h represents the embedding dimension (e.g., 128), all adjacent vehicle embedding feature is $\mathbf{Bv}_i \in \mathbb{R}^{3 \times h}$.

The Adj-pedestrian embedding module is designed similarly to the Adj-vehicle embedding. It uses the adjacent pedestrian features $\mathbf{Ap} \in \mathbb{R}^{3 \times 3}$, each adjacent pedestrian feature denoted as $\mathbf{Ap}_j \in \mathbb{R}^3$, where $j \in \{0, 1, 2\}$. The resulting embedding is denoted as $\mathbf{Bp}_j$, and the process is as follows:

$$\mathbf{Bp}_j = \mathbf{W}_{pd}(\mathbf{Ap}_j) + \mathbf{W}_{ps}(\mathbf{Ap}_j) \quad (2)$$

where $\mathbf{W}_{pd}$ and $\mathbf{W}_{ps}$ correspond to multilayer and single-layer linear transformations, respectively. The dimension of the generated $\mathbf{Bp}_j$ is $\mathbb{R}^h$ and adjacent pedestrian embedding feature is $\mathbf{Bp} \in \mathbb{R}^{3 \times h}$.

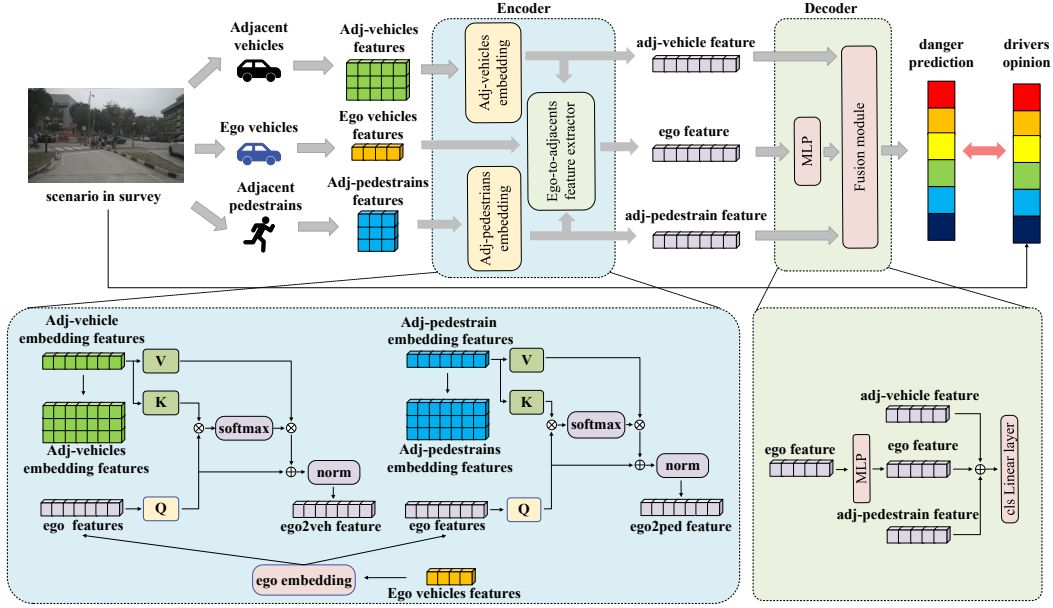


Fig. 2: Framework of the RPT. The driving scenes are extracted from the NuScenes dataset and provided in the questionnaire with hidden states of vehicles and pedestrians. The survey answers are processed to be concatenated with the hidden states and then embedded to predict the risk level. The Transformer-based encoder extracts features, which are then processed by a decoder composed of multiple linear layers to generate danger predictions. Both the danger predictions and driver options are fed into the agent to assist decision-making.

2) *Ego-to-adjacents feature extractor*: In this module, we will get ego embedding feature, ego2veh feature and ego2ped feature. Here we consider adjacent vehicle i and adjacent pedestrian j . Ego embedding feature is denoted as $\mathbf{E} \in \mathbb{R}^4$, which stacks ego's velocity, the number of vehicles and the number of pedestrians in scenario within 150 meters. We first embed ego embedding feature as $\mathbf{E}_{eb} = \mathbf{W}_{eb}(\mathbf{E})$, where $\mathbf{W}_{eb}$ is learnable linear layer, and $\mathbf{E}_{eb} \in \mathbb{R}^h$.

Due to the cross-attention mechanism's ability to enhance the model's feature extraction by learning weighted responses across different scales, it effectively extracts key features from multi-modal inputs during feature fusion [26], [27], [28]. Therefore, we employ the cross-attention mechanism to extract and fuse features for both adjacent vehicles and pedestrians, resulting in the ego-to-adjacent pedestrian and ego-to-adjacent vehicle features.

For a adjacent vehicle i , we employ cross-attention mechanism and perform the following operations with the ego vehicle:

$$\mathbf{F}_{ego2veh} = \text{Attn}(\mathbf{Q}_{e2v}(\mathbf{E}_{eb}), \mathbf{K}_{e2v}(\mathbf{Bv}_i), \mathbf{V}_{e2v}(\mathbf{Bv}_i)) \quad (3)$$

$$\mathbf{F}_{ego2veh} = \text{norm}(\mathbf{F}_{ego2veh} + \mathbf{Q}_{e2v}(\mathbf{E}_{eb})) \quad (4)$$

$$\mathbf{F}_{ego2veh} = \text{norm}(\text{FFN}(\mathbf{F}_{ego2veh}) + \mathbf{F}_{ego2veh}) \quad (5)$$

$\mathbf{Q}_{e2v}(\cdot)$, $\mathbf{K}_{e2v}(\cdot)$, and $\mathbf{V}_{e2v}(\cdot)$ represent the embedding operations, $\text{Attn}(\cdot)$ denotes the attention mechanism, and $\text{FFN}(\cdot)$ is the feed-forward network (a multi-layer perceptron).

For adjacent pedestrian j , we applied a method similar to the fusion of adjacent vehicles and the ego to fuse the features of adjacent pedestrians and the ego. We first get

attention weight as follow:

$$\mathbf{F}_{ego2ped} = \text{Attn}(\mathbf{Q}_{e2p}(\mathbf{E}_{eb}), \mathbf{K}_{e2p}(\mathbf{Bp}_j), \mathbf{V}_{e2p}(\mathbf{Bp}_j)) \quad (6)$$

$\mathbf{Q}_{e2p}(\cdot)$, $\mathbf{K}_{e2p}(\cdot)$, and $\mathbf{V}_{e2p}(\cdot)$ are the embedding operations for the pedestrian j feature. $\mathbf{F}_{ego2ped}$ will then be processed by norm and FFN, which are same with the $\mathbf{F}_{ego2veh}$ processing. Finally, the output of Ego-to-Adjacents feature extractor is $\mathbf{F}_{ego} \in \mathbb{R}^h$, which fuse $\mathbf{E}_{eb}$, $\mathbf{F}_{ego2veh}$ and $\mathbf{F}_{ego2ped}$ by:

$$\mathbf{F}_{ego} = \text{sum}(\mathbf{E}_{eb}, \mathbf{F}_{ego2veh}, \mathbf{F}_{ego2ped}) \quad (7)$$

3) *Decoder*: The decoder is responsible for fusing the features from neighboring vehicles $\mathbf{Bv}$, pedestrians $\mathbf{Bp}$, and the ego features $\mathbf{F}_{ego}$ to predict risk. In decoder, $\mathbf{F}_{ego}$ is updated by two-layer Multi-Layer Perceptron (MLP). $\mathbf{Bv}$ and $\mathbf{Bp}$ are updated by $\text{mean}(\cdot)$ to gets the mean value of all $\mathbf{Bv}_i$ and $\mathbf{Bp}_j$, $i \in \{0, 1, 2\}$, $j \in \{0, 1, 2\}$. All three features are fused in three-layers MLP as follow:

$$\mathbf{F}_{adj} = \text{MLP}(\mathbf{Bv} + \mathbf{Bp} + \mathbf{F}_{ego}) \quad (8)$$

Finally, the risk prediction is obtained by applying a softmax function to the output of a learnable linear layer $\mathbf{W}_r$ that operates on the concatenation of $\mathbf{F}_{ego}$ and $\mathbf{F}_{adj}$:

$$\text{risk prediction} = \text{softmax}(\mathbf{W}_r(\text{concat}(\mathbf{F}_{ego}, \mathbf{F}_{adj}))) \quad (9)$$

Here, $\mathbf{W}_r \in \mathbb{R}^{(2 \times h) \times 2}$ is also a learnable linear layer. The risk prediction output $\text{risk_pred} \in \mathbb{R}^6$ is designed to fit the statistical results of the risk assessment from a survey.

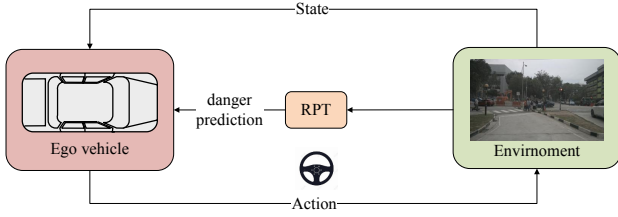


Fig. 3: Training process of GORA-RL. Here, we use RPT to estimate the level of risk in a given scenario to train the PPO.

4) *Training loss*: The RPT model produces a set of probability vectors to fit the statistical data collected from the survey, which reflects the drivers' assessment of risk levels in various scenarios. To optimize the RPT model, the Mean Squared Error (MSE) loss function is used to supervise the learning process by measuring the difference between the model's predictions and the actual survey data. The loss function is formulated as follows:

$$\mathcal{L}_{\text{RPT}} = \mathcal{L}_{\text{mse}}(\text{risk_pred}, \text{survey_prob}) \quad (10)$$

where *risk_pred* represents the risk predictions output by the RPT model, and *survey_prob* denotes the probabilities derived from the survey data.

C. GORA-RL

We use the RPT as the reward function to train the proposed GORA-RL approach, as shown in Fig. 3. Typically, driving decisions are formulated as a Partially Observable Markov Decision Process (POMDP) $\langle \mathcal{S}, \mathcal{O}, \mathcal{A}, \mathcal{R}, \mathcal{P}, \gamma \rangle$, where $\mathcal{S}$ is the environment states such as positions and velocities of vehicles/pedestrians, $\mathcal{O}$ is the observations from the ego vehicle's sensors, and $a_t \in \mathcal{A}$ is actions taken (throttle, brake, steering) at the current state $s_t \in \mathcal{S}$ to the next state s_{t+1} and obtain the reward $r_t \in \mathcal{R}$ with the transition probabilities $\mathcal{P}$. γ is the discount factor.

1) *Environment*: We simulate a driving environment with multiple vehicles and pedestrians interacting in real-time using SMARTS [29]. The environment is based on realistic driving conditions and includes dynamic elements such as neighboring cars, pedestrians, and road constraints. Each vehicle is assigned a mission that specifies its destination, and the ego vehicle's objective is to reach this destination while avoiding collisions and minimizing risk. The environment allows for multiple autonomous agents (vehicles) to navigate simultaneously. It provides full access to traffic information, lane-specific data, and pedestrian locations. Ego vehicles and their neighboring agents are equipped with sensors that continuously monitor their surroundings, including the positions and velocities of nearby objects. In this work, we trained over four scenarios, namely, straight scenario, loop scenario, merge scenario, and T-intersection scenario, as illustrated in Fig. 4.

2) *Observation Space*: The observation space provides the ego vehicle with comprehensive data necessary for

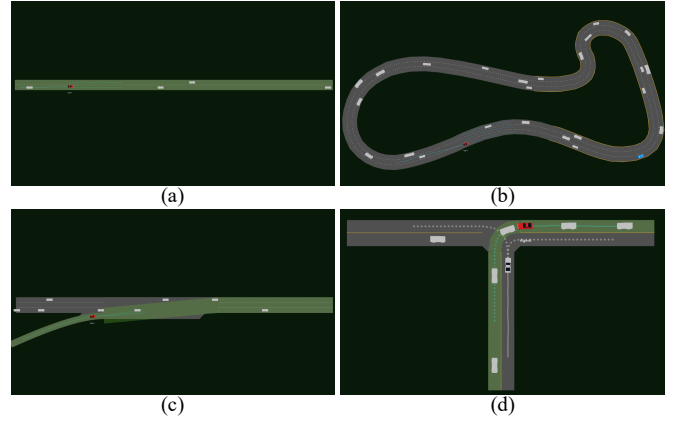


Fig. 4: The driving scenarios used in SMARTS, where the red car is the ego vehicle while the gray vehicles are the adjacent vehicles. (a) Straight scenario; (b) Loop scenario; (c) Merge scenario; (d) T-intersection scenario.

decision-making, capturing both its own state and interactions with the environment. The observations used include, **ego-position** $P_{ego} \in \mathbb{R}^3$ (Center coordinates of the vehicle's bounding box's bottom plane), **bounding_box** $bbx \in \mathbb{R}^{k \times 3}$ (Dimensions (length, width, height) of the vehicles), **heading** $\psi_{ego} \in \mathbb{R}$ (Vehicle's facing direction (radians)), **speed** $\|V\|_{ego} \in \mathbb{R}$ (Travel speed (m/s)), **steering** $\theta_{ego} \in [-\pi, \pi]$ (Front wheel angle (radians)), **yaw_rate** $\dot{\psi}_{ego} \in \mathbb{R}$ (Rate of heading rotation about the z-axis (rad/s)), **road_id**, **lane_id** (Identifiers for the nearest road and lane), **lane_index** (Index of the nearest lane (rightmost lane = 0)), **mission** (Vehicle's desired destination), **linear_velocity** $V_{ego} \in \mathbb{R}^3$ (Vehicle velocities in the global coordinate frame). This observation space provides the ego vehicle with rich data about its own state, its position relative to the road and neighboring vehicles, and other dynamics critical for decision-making. These observations are also selected to generate the risk level with the RPT model.

3) *Action Space*: The action space consists of continuous control inputs that allow the vehicle to adjust its movement in the environment. The actions are represented as a tuple containing three parameters, namely **throttle**: a continuous value in the range $[0, 1]$, where 0 represents no acceleration and 1 represents full acceleration; **brake**: a continuous value in the range $[0, 1]$, where 0 represents no braking and 1 represents full braking; and **steering**: a continuous value in the range $[-1, 1]$, where -1 corresponds to a full left turn, 0 is neutral (straight), and 1 corresponds to a full right turn. This action space allows the ego vehicle to adjust its speed and direction, ensuring flexible control for responding to varying traffic conditions and dynamic obstacles.

4) *Reward Space*: The reward function is central to shaping the behavior of the autonomous agent. In this framework, the base reward is derived from the distance traveled by the vehicle. The reward function is defined as:

$$R_{\text{base}}(x) = \begin{cases} x & \text{if } x > 0.5 \\ 0 & \text{otherwise} \end{cases} \quad (11)$$

where x (unit: m) represents the distance traveled by the vehicle in a given time step. If the distance traveled is greater than 0.5 meters, the agent is rewarded proportionally; otherwise, the reward is zero. In addition to the base reward, the risk-aware reward function incorporates a penalty for unsafe behavior. This is achieved by mapping the risk levels obtained from the RPT to corresponding reward values. In this work, $R_{\text{risk}}(r) = r$, where r is the predicted risk level. Higher risk levels result in greater penalties, thereby encouraging the agent to take actions that minimize the risk while still achieving the mission goal. The final reward function is formulated as:

$$R_{\text{total}}(x, r) = R_{\text{base}}(x) - \lambda \cdot R_{\text{risk}}(r) \quad (12)$$

where λ is a scaling factor that balances the trade-off between maximizing the distance traveled and minimizing risk. The proposed reward structure enables the ego vehicle to prioritize both safety and efficiency, ensuring it can make risk-aware decisions while navigating complex driving environments. Here, $\lambda = 1.0$ is set in this work. If the vehicle collides with others, drives off road or off route, then the episode ends.

5) *Training Scheme*: PPO is chosen to train the ego vehicle's policy, for its stability and sample efficiency in continuous control tasks, which are critical in autonomous driving. The key components are as follows:

- **Policy Update**: PPO performs clipped updates to the policy to prevent overly large changes, ensuring stable training. The objective function maximizes the expected reward while penalizing updates that deviate too far from the previous policy, thus maintaining a balance between exploration and exploitation.
- **Actor-Critic Framework**: The PPO algorithm uses an actor-critic framework where the policy network (actor) proposes actions based on the observation space, and a value network (critic) estimates the value function to guide learning. The critic helps evaluate the expected future rewards given the current policy, reducing variance during policy updates.
- **Risk-Aware Reward Function**: The PPO algorithm optimizes the risk-enhanced reward function $R_{\text{total}}(x, r)$, which incorporates both the base reward for distance traveled and penalties based on the predicted risk level. This encourages the policy to learn safer driving behaviors while maintaining travel mission performance.
- **Training Iterations**: Training proceeds in multiple episodes, where the ego vehicle interacts with the environment, collects trajectory data, and updates the policy using PPO. Each episode includes varying traffic conditions and road layouts, allowing the agent to generalize its learned policy across diverse driving scenarios.

Our training process continues until the policy converges, leading to improved success rates and reduced risk in decision-making.

TABLE II: Training parameters for GORA-RL

Parameter	Value	Parameter	Value
batch size	2000	minibatch size	128
KL coefficient	0.2	KL target	0.01
discount factor	0.99	clip value	0.3
sgd iteration	30	maximum episode steps	500

IV. EXPERIMENTS AND RESULTS

This section introduces the experimental setup, results, and discussion.

A. Experiments and Settings

1) *Settings for RPT*: Since there are no existing models that use key scene information for scenario risk prediction, we evaluate the efficiency of the RPT model using a baseline model composed entirely of linear layers. The evaluation metrics include the number of model parameters, validation loss, and Floating Point Operations (FLOPs). The baseline method's encoder and decoder are built with a five-layer Multilayer Perceptron (MLP), where the hidden layer dimension is 256. In the RPT model, the hidden layer dimension is 128, and the number of attention heads is 8. Both the baseline and RPT models are trained on a single NVIDIA 3070Ti GPU with a batch size of 1 for 300 epochs. The AdamW [30] optimizer is used with a learning rate of 1×10^{-5} and a weight decay of 1×10^{-6} .

2) *Settings for GORA-RL*: We consider two SOTA RL baselines: PPO without RPT [15] (PPO), and a SOTA risk-aware RL method, the risk-aware DQN (RADQN) [13], which uses heuristic risk rewards based on the conflict zone. All algorithms are trained using the SMART environment and a 3070Ti GPU. The learning rate is a linear scheduler from (1e-3, 1e-5). A two-layer fully connected NN (FCN) of size 256 is used for the actor network and value network with tanh as the activation function. The detailed parameters are listed in Table II. Three random runs of training are performed for each mission to minimize the effects of randomness from a single run. The comparison metric is the reward, which represents the average traveling distance until termination. It is also the core metric used to evaluate the performance of the AV.

B. Results and Discussion

The risk prediction performance is listed in Table III. Compared with the baseline model composed entirely of linear layers, the RPT model achieves higher prediction accuracy on the validation dataset. Additionally, the number of parameters in the RPT model is approximately half that of the linear baseline model. The FLOPs per scene are also lower for the RPT model, demonstrating faster inference speed for individual scenes. The reward curves of various RL methods are shown in Fig. 5. With the RPT module, our GORA-RL algorithm converges to a higher mean reward in all four scenarios compared to the baseline methods. Specifically, our method achieves significantly better mean rewards (i.e., longer traveling distances: 279.90m and 216.79m) in the merge and intersection scenarios, which are the most

TABLE III: Comparison of parameters, valid losses, and FLOPs for different risk prediction models.

Model	Parameters	Valid Loss	FLOPs (GFLOPs)
Baseline	2.69062×10^5	2.01×10^{-2}	7.94120×10^{-4}
RPT	1.29944×10^5	8.8×10^{-3}	2.72760×10^{-4}

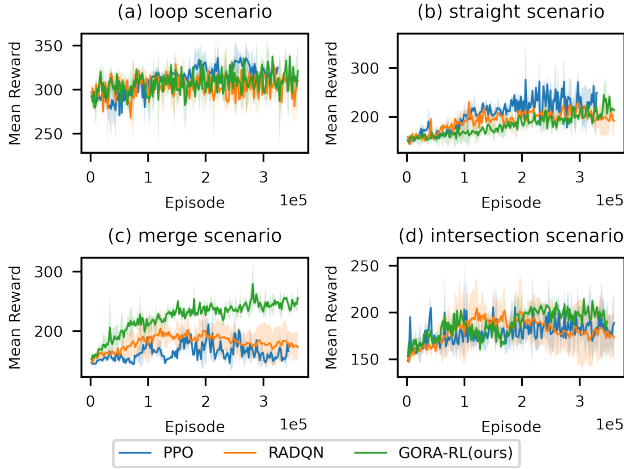


Fig. 5: Comparison with baseline methods in terms of reward. Here, bold results represent the mean reward curves from three random runs, and the shaded area represents the corresponding standard deviation.

risky, compared to the baseline methods. This validates the advantage of group opinion risk prediction for ensuring safe AV behavior in cluttered environments.

V. CONCLUSION

This work develops a novel risk-aware RL method, referred to as GORA-RL, for safety decision-making in autonomous driving. The method first trains a transformer-based risk prediction model using survey data from multiple expert human drivers on risk assessments and preferred driving behaviors. This model is then used as a reward function to guide the training of PPO, which is employed to determine the vehicle's driving behavior. Experimental results show that our GORA-RL outperforms two SOTA RL methods in several complex driving scenarios, in terms of reward and other metrics. Future work will explore applying the proposed method to real vehicles.

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